Overall Design Structure

One continuous spine × Five urban relation transects × Design response windows for three key areas

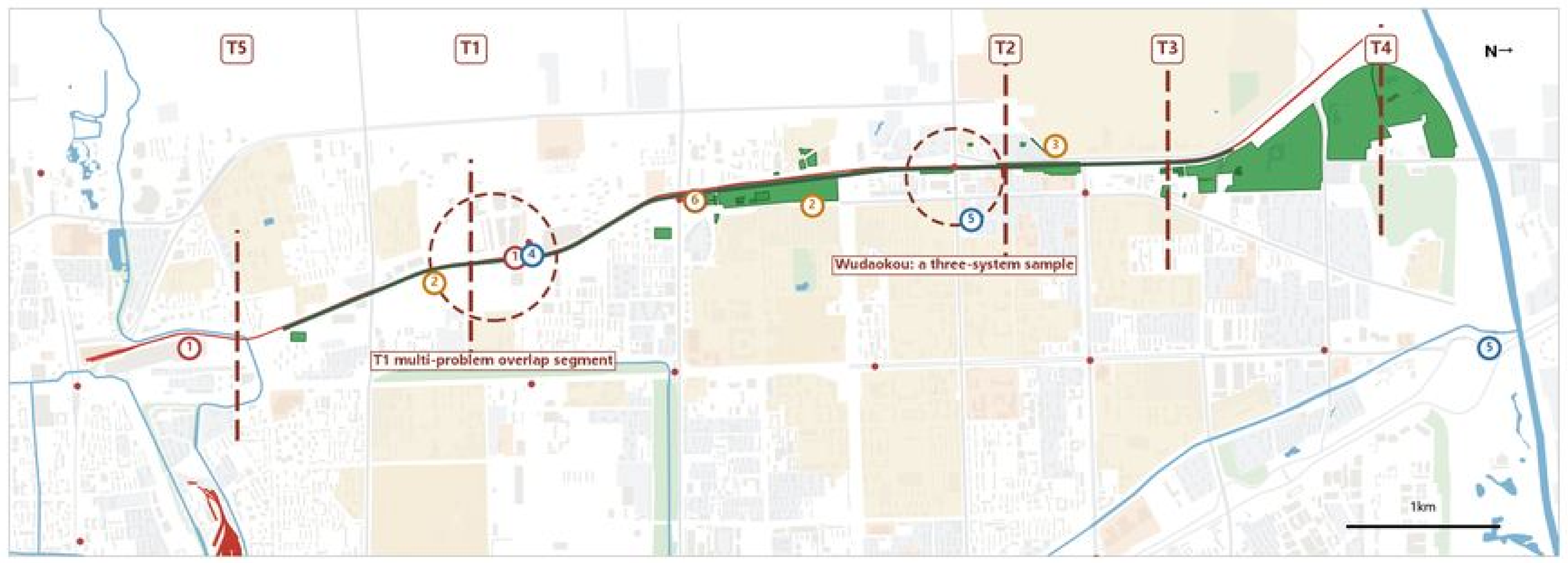
The Jing-Zhang public space spine is now continuous; the lateral public relationship with the cities on either side is not.

Not five points along the line, but a reorganisation of the relationship between one spine and the city: five transects identify the recurring problems, and the response windows for the three key areas carry the focused response.

- Jing-Zhang spine (primary)
- Response window (proposal-defined; not the official key area)
- T1 Station–City Reorganisation (DIRECT)
- T2 Proximity → Exchange (FACE+USE)
- T3 Boundary → Exchange Interface (FACE+CROSS)
- T4 Rear Edge → Frontage (FACE+USE)
- T5 Terminus → Gateway (ARRIVE+CROSS)
- Ordinary segments: keep existing conditions
- Coordination nodes (three)
- Crossing · walkable (NOW)
- Crossing · TO VERIFY (CONDITIONAL)
- North Third Ring bridge zone
- Arrival arm ARRIVE
- Interface arm FACE (controlled)
- T5 east–west via the station area
- Conditional green-network interface (T4)
- Line 13A interface RESERVE
- Indicative task area · not official
- Relation transects T1–T5

The three rings on the plan are the response windows identified by this proposal along Jing-Zhang; not equivalent to the official key-area extent.

Three-System Synthesis | GRAY × SOCIAL × GREEN



The three infrastructure deficits overlap in different combinations, producing a different urban relation problem at each of the five transects T1–T5.

Five Urban Relation Transects | Index

T1 **Station–City Relationship Reorganisation** — ARRIVE FACE CROSS
Dazhongsi

T2 **Proximity → Exchange** — FACE USE
Chengfu Road–Wudaokou

T3 **Boundary → Conditional Exchange Interface** — FACE CROSS
Tsinghua East–Xueyuan Road

T4 **Industrial Rear Edge → Urban Frontage** — FACE USE
Zhongzhiyuan

T5 **Railway Terminus → Urban Gateway** — ARRIVE CROSS
Beijing North Railway Station

Transects are design units, not zoning districts; T1 is taken to object level on the Key Areas board.

Source: urban relation analysis and design work of this proposal; base data and evidence records are held in the submission package.

A0-02 Land Use and Urban Renewal

SOCIAL | functional organisation · public-space exchange · renewal of built space

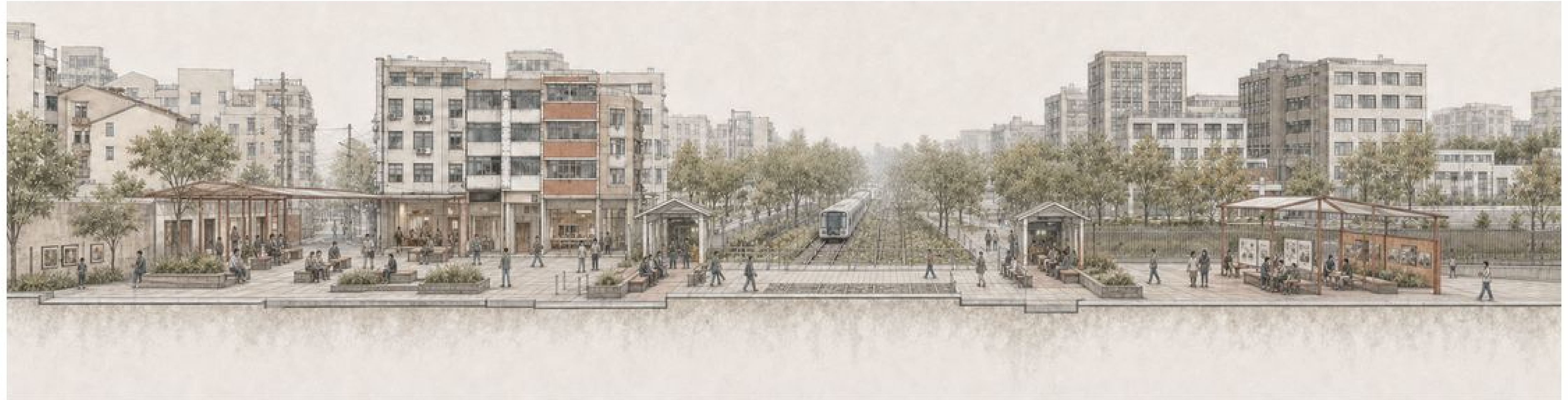
Renewal is not about redrawing parcels, but about reorganising functions, interfaces and public exchange relationships.

Assuming neither large-scale demolition nor statutory land-use change, this board answers how innovation exchange can be built inside the existing urban fabric through functional organisation, public space, ground-floor interfaces and operation.

T2 | Proximity → Exchange

Chengfu Road–Wudaokou FACE USE

EXISTING URBAN RELATIONS



FUTURE RELATION SCENARIO

Existing | community — passing by — existing crossing — passing by — universities / innovation resources Future | community exchange — arrival across the corridor — managed innovation exchange

T3 | Boundary → Conditional Exchange Interface

Tsinghua East–Xueyuan Road FACE CROSS

EXISTING URBAN RELATIONS

FUTURE RELATION SCENARIO

Existing | a single hard boundary Future | layered, controlled exchange interface: closed but publicly oriented — semi-open exchange — conditionally controlled opening; opening locations are design-defined and do not correspond to existing campus gates.

Four Renewal Modes

- **DIRECT DESIGN**
  public space, ground-floor interfaces, arrival organisation
- **PLANNING CONTROL**
  interface control guidelines, conditional openings
- **MANAGEMENT / OPERATION**
  time-shared use, opening hours, programme organisation
- **KEEP / NO DESIGN**
  no fence removal, no forced opening — also a planning decision

All four modes coexist within the same block; which mode applies where is itself part of the planning decision.

SOCIAL evidence | resources are clustered; exchange interfaces and openness are not continuous

Campus boundary | detour length is set by gate location.

SOCIAL system structure | universities / innovation resources · communities / everyday services · functional relations in the built area.

In a built-up area, what can be designed is the relationship, not the parcel.

Resources already clustered
Universities, research, translation services, industrial parks and everyday services are densely clustered along the corridor.

Weak exchange interface
Public space is the most direct potential interface between them, and it is the weakest layer.

The mismatch is in real walking
Facilities are not lacking; access, openness and exchange conditions are.

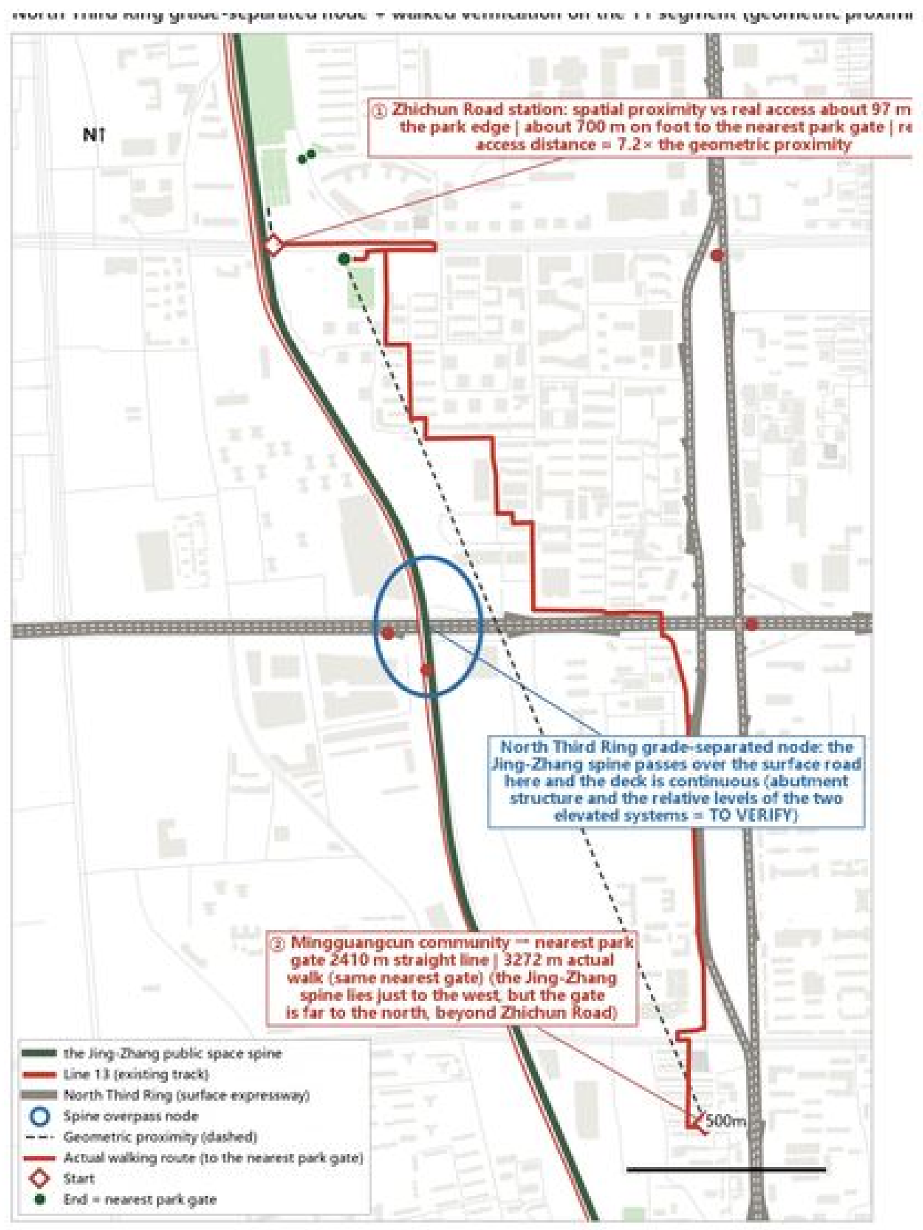
A Station ≠ Real Access. Spatial Proximity ≠ Easy Access.

The transport infrastructure that supports north–south operation also forms the lateral barrier to east–west urban relations. No new rail crossing is proposed; the response is organised through station areas, arrival and gateway relationships.

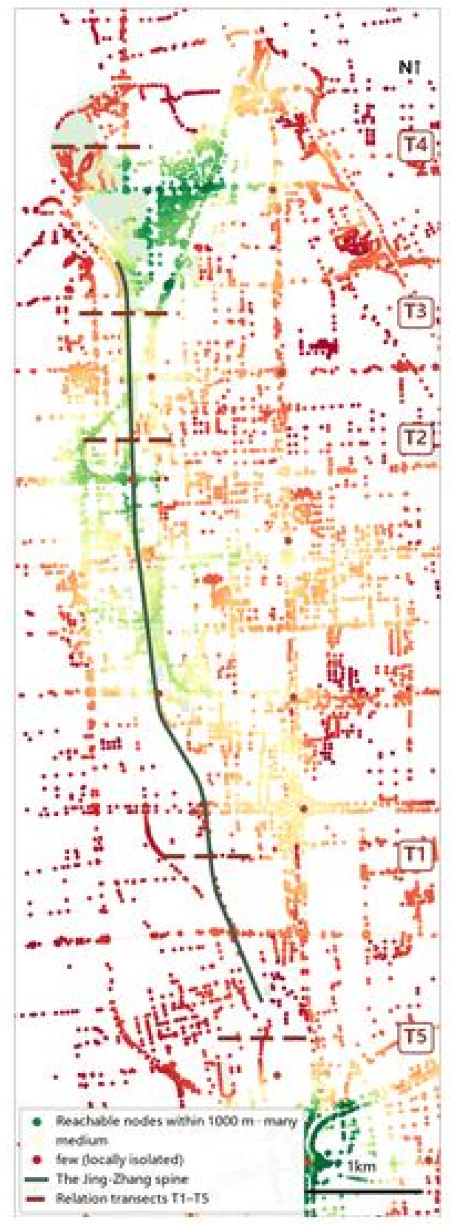
GRAY system | operating spine and lateral barrier

Jing-Zhang alignment (spine) · Line 13 + operating band · High-speed at-grade / yard tracks · Beijing North rail yard · Military area · Expressway / arterial · Crossing · walkable (3) · Crossing · TO VERIFY (3) · North Third Ring overpass · Relation transects T1–T5

The Line 13 shared corridor, the North Third and Fourth Ring expressways, the Beijing North rail yard and the at-grade high-speed sections form the north–south operating network and, at the same time, the principal east–west barrier; the seven existing crossings fall into three classes by level of verification.



North Third Ring grade-separated node and walked verification on the T1 segment | Zhichun Road station is about 97 m from the park edge but about 700 m on foot from the nearest gate; the Mingguangcun resettlement community is 2,410 m in a straight line and 3,272 m on foot.



Local connectivity | consistently low around university compounds and rail yards.

It looks close. Why can you not get there?

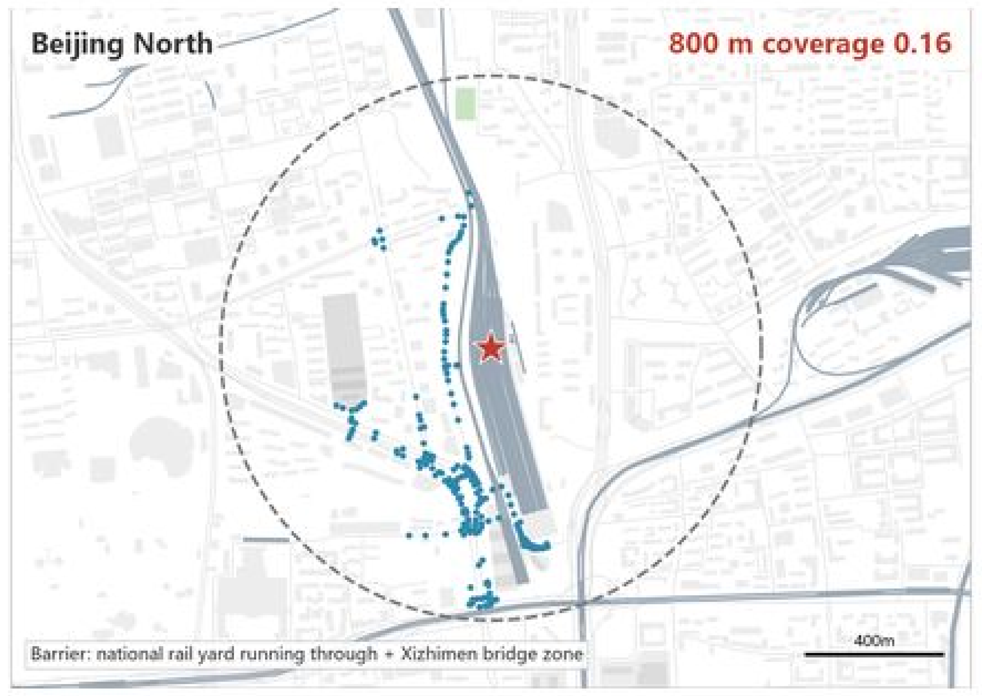
Crossings exist at only a few points and are mostly purely traffic structures. The gap between spatial proximity and real access is the principal relation problem along this stretch.

Utility support

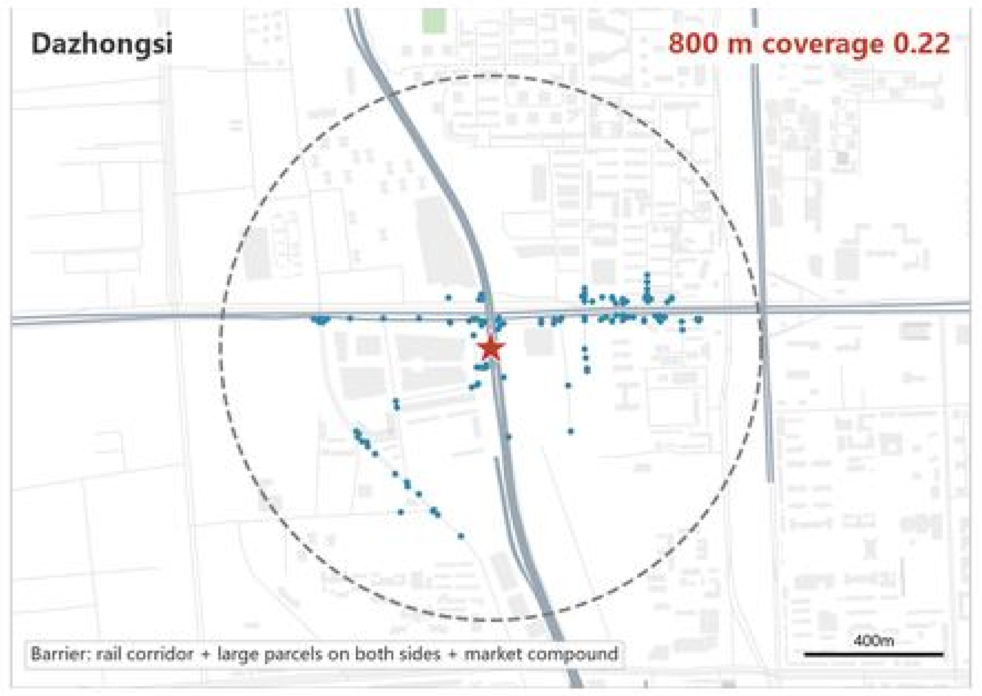
- Space reserved for environmental sensing
- Continuous path for low-speed devices
- Public information interface

Only conceptual support with present evidence is shown; no pipework, capacity or engineering design is proposed.

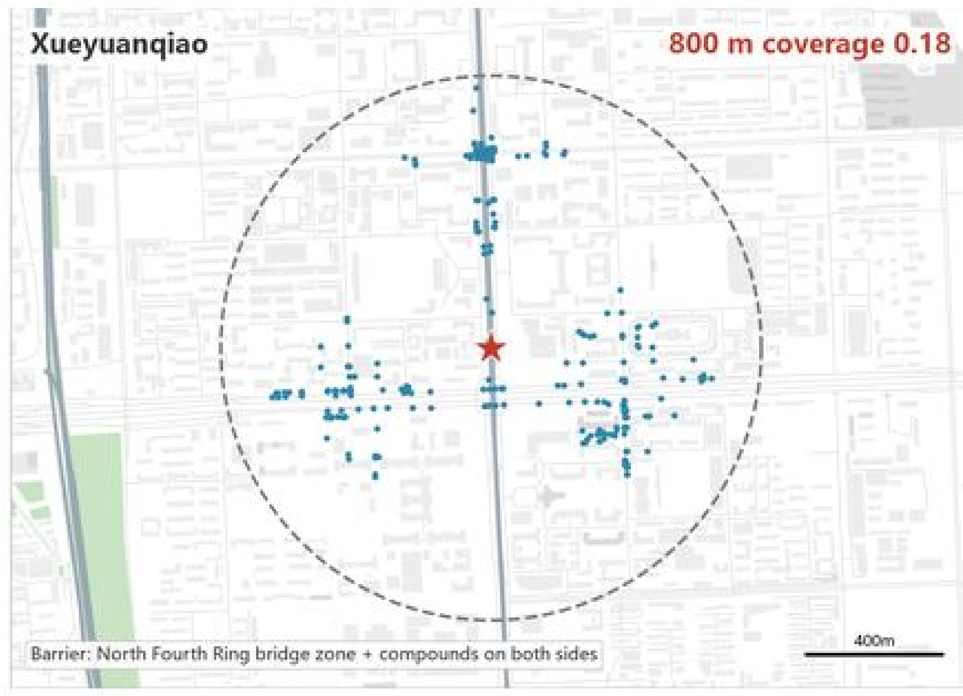
A Station ≠ Real Access | the gap between the 800 m standard catchment and the real walking network



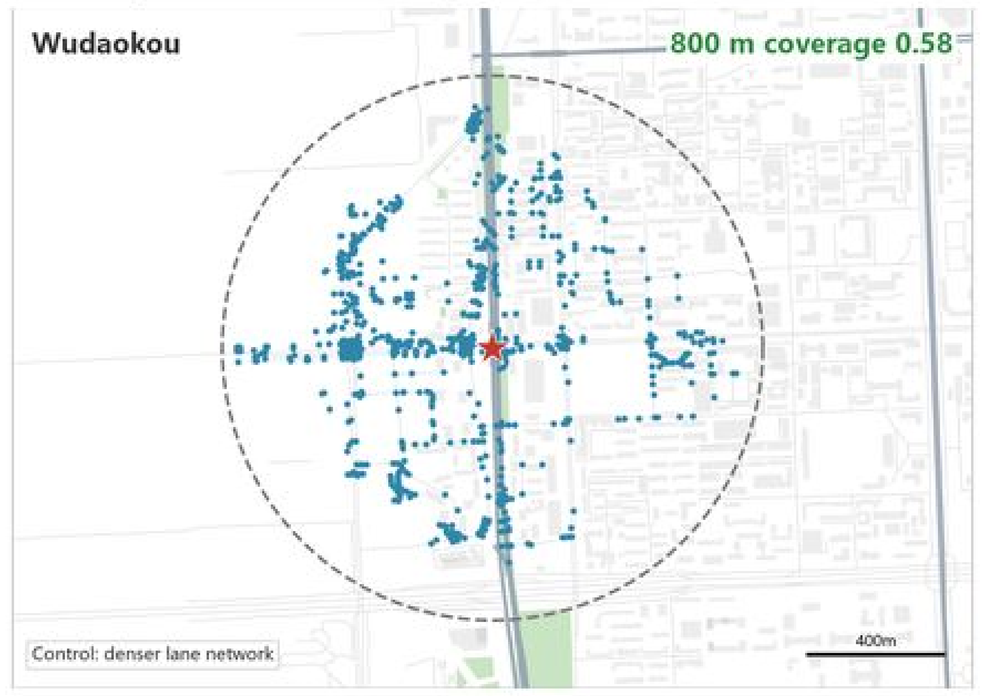
01 Beijing North | coverage 0.16



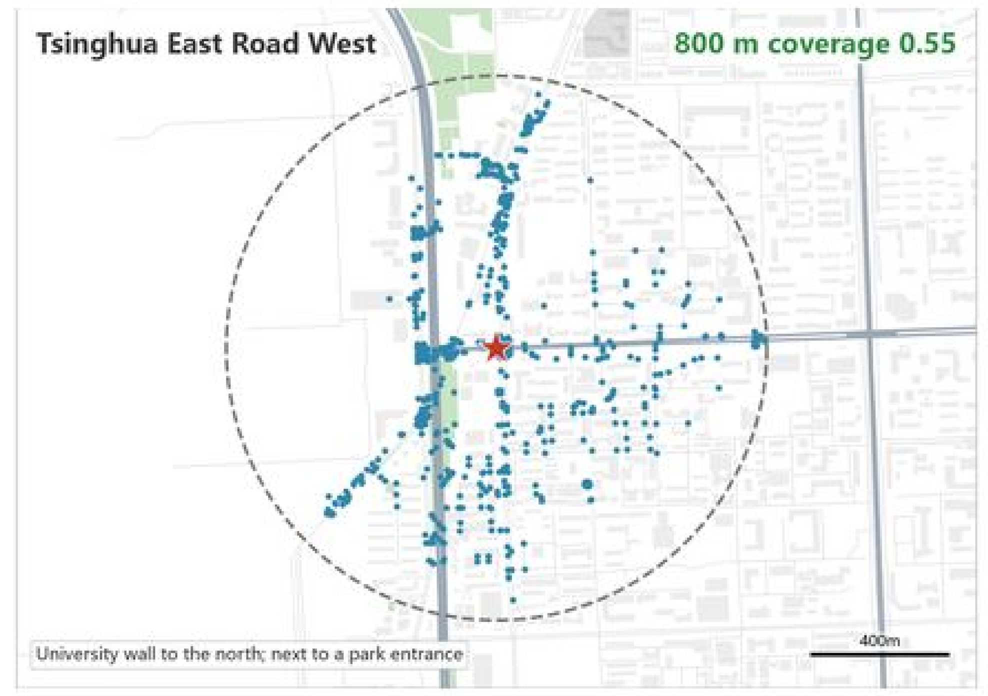
02 Dazhongsi | coverage 0.22



03 Xueyuanqiao | coverage 0.18



04 Wudaokou | coverage 0.58 (control)



05 Tsinghua East Road West | coverage 0.55 (next to a park gate)

T5 | Railway Terminus → Urban Gateway

Beijing North Railway Station ARRIVE CROSS

EXISTING URBAN RELATIONS

FUTURE RELATION SCENARIO

The rail yard is both an operating system and a barrier.

East and west are linked through the station area, arrival and gateway relationships. No new rail crossing is proposed at this stage.

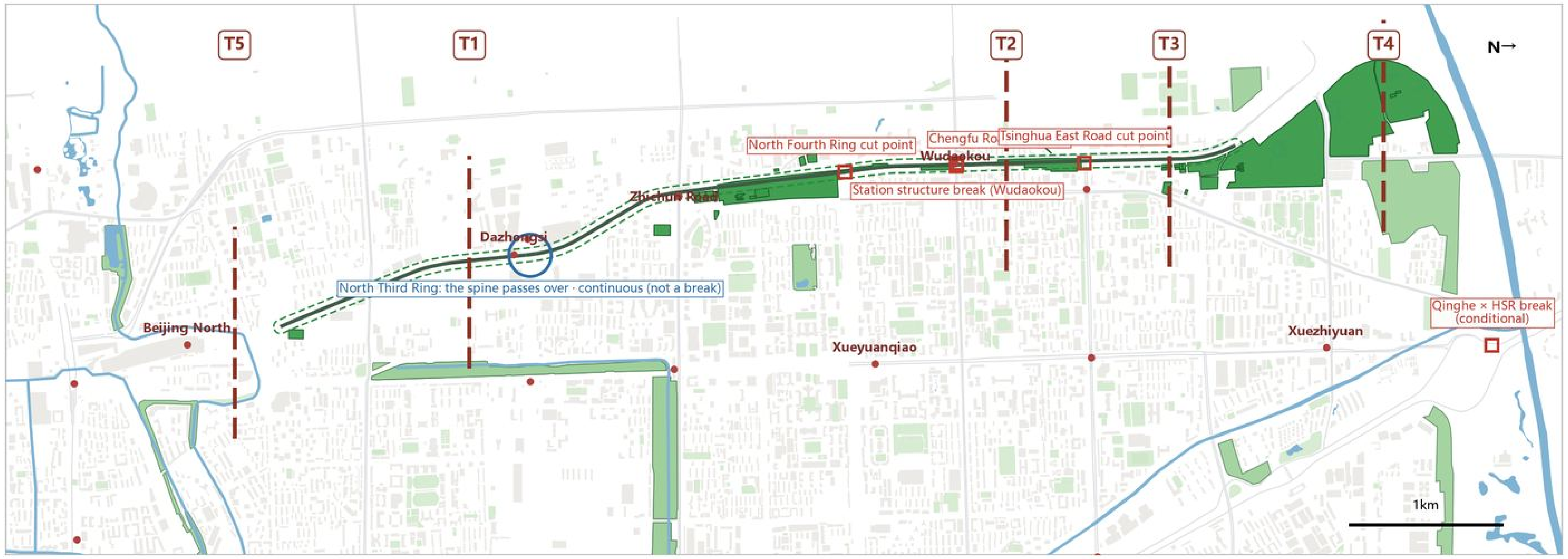
Jing-Zhang Railway Heritage Park and Blue–Green Public Space

GREEN | green spine · public entry · continuity of use

Ecological Continuity ≠ Continuous Public Use.

Green space is not background planting but the medium through which industry, everyday life and Jing-Zhang form a public relationship. The point is not more green, but turning the ecological spine into space that can be entered, reached and used.

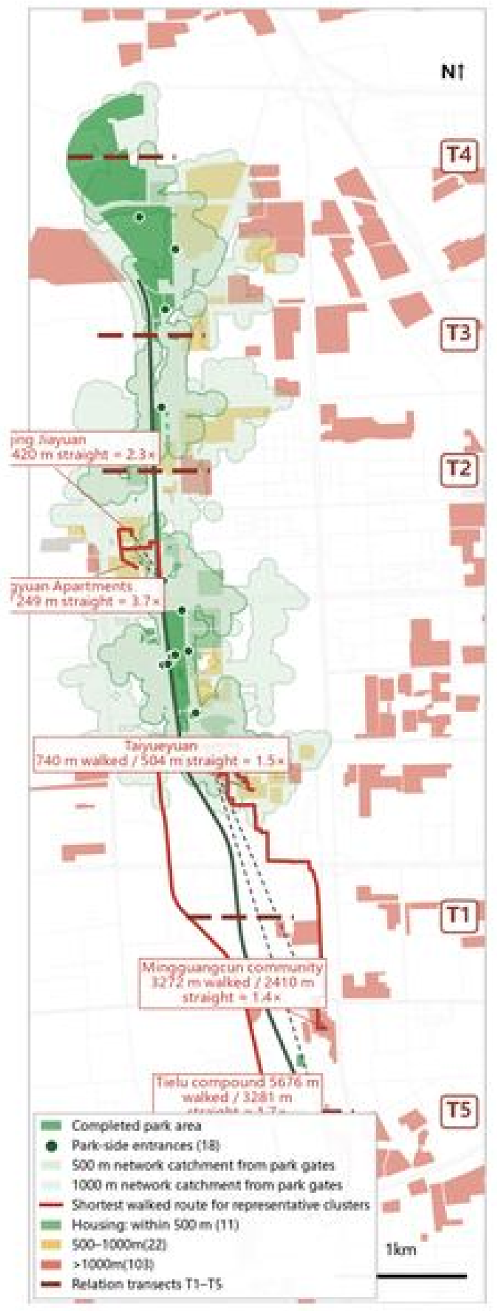
GREEN system | green spine and structural breaks



Completed section (CORRIDOR) | Green patches (> 3 ha) | Water | North Third Ring · continuous
Spine band (phase two) | Other green space | Structural breaks (5) | Relation transects T1–T5

Jing-Zhang is one of the principal continuous green public-space spines within the study area. There are five structural breaks, four of them caused by grey infrastructure; at the North Third Ring the spine passes over and the green corridor stays continuous.

Ecological / physical continuity



Residential areas → real access to Jing-Zhang public space

Green on the Map ≠ Structural Connection

In the northern section green quantity is large and adjacent and the blue line is continuous; the ecological base is not the main problem.

Five structural breaks
Four of them are caused by grey infrastructure.

Adjacent but hard to reach
Large residential areas beyond the rail and the expressway lie more than 1,000 m away.

Entrances decide use
Walked / straight-line ratio 1.4–3.7×: gate location matters more than green quantity.

T4 | Industrial Rear Edge → Urban Frontage

Zhongzhiyuan FACE USE

EXISTING URBAN RELATIONS

FUTURE RELATION SCENARIO

Industrial operation → Public ground floor → Semi-outdoor exchange → Public forecourt / everyday green → Jing-Zhang → Urban life

CONDITIONAL GREEN-NETWORK INTERFACE (northern end–Qinghe) is a secondary annotation only: reserved solely if an existing lawful crossing condition holds, and not raised to a primary T4 mechanism.

The three areas carry different tasks and therefore different design depths: only Dazhongsi is taken to object level.

A TASK AREA is the extent the competition asks us to address; a RESPONSE WINDOW is the relationship this proposal actually reorganises within it. They are not the same extent.

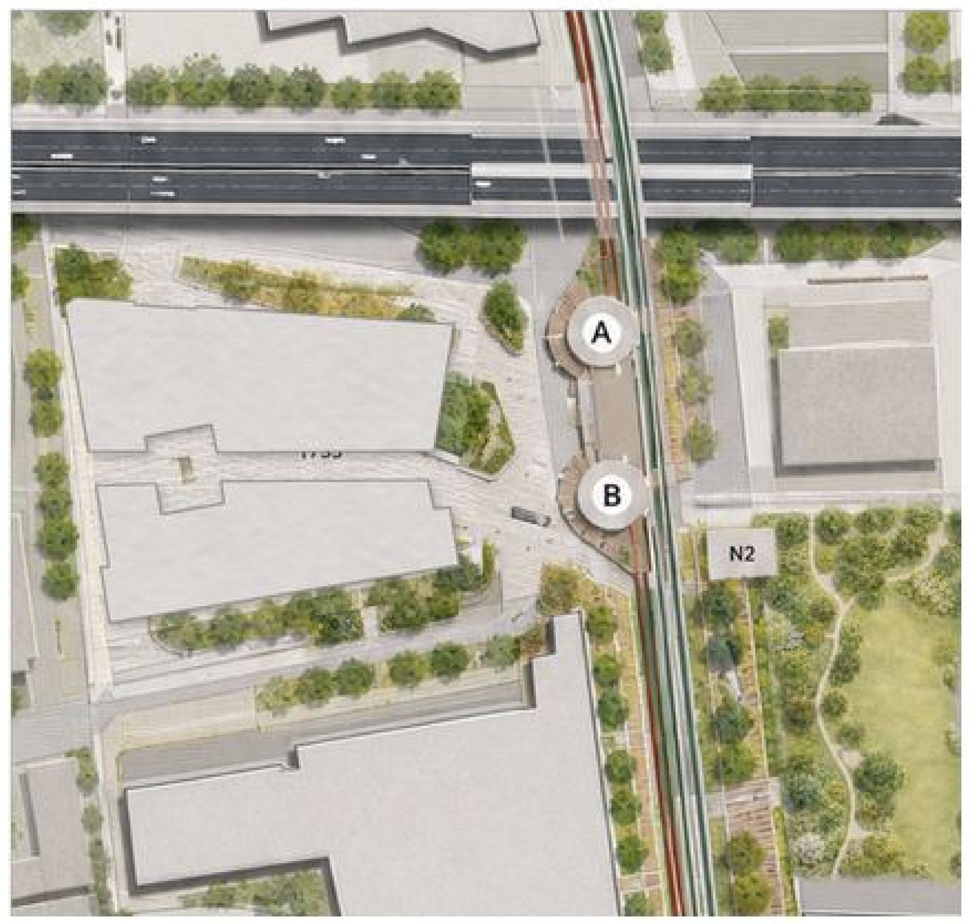
A | Dazhongsi AI Industry Cluster

RESPONSE WINDOW T1 Dazhongsi station area | Station–City Relationship Reorganisation

OBJECT-LEVEL DESIGN

ARRIVE FACE CROSS

KEY DESIGN SECTION | GROUND-LEVEL ARRIVAL → TWO-LEVEL PUBLIC HINGE WITHIN THE STATION → UPPER-LEVEL PUBLIC TRANSFER → LINE 13 PLATFORM + JING-ZHANG RAILWAY HERITAGE PARK.



CORE NODE PARTIAL PLAN

CORE NODE AERIAL VIEW

Core design moves

01 OPEN

City approach → the 1733 forecourt → Entrance A / Entrance B

02 HINGE

A two-level public transfer hinge within the station; no change to rail engineering

03 RELEASE

Released towards the Jing-Zhang spine, N2, the eastern public space and the city on both sides

VERIFIED: plan-level factual relations between 1733, Entrances A/B, the station structure, Line 13, Jing-Zhang and the North Third Ring.

TO VERIFY: exact levels, vertical circulation within the station structure, engineering interfaces, landing conditions of N2.

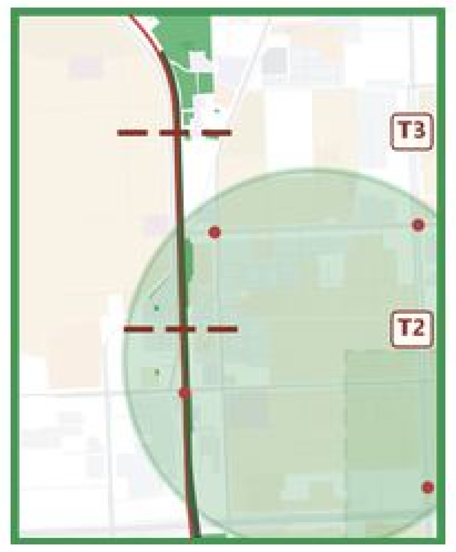
NOT PROPOSED AT THIS STAGE: any unverified new rail crossing.

Official task objects | location key

Task-area extents are indicative (provisional), not official redlines; areas are the published figures.

B | Beijing AI Origin Community

About 104.3 ha | world-class AI innovation ecosystem RESPONSE WINDOW T2 (adjacent condition: the T3 campus boundary)

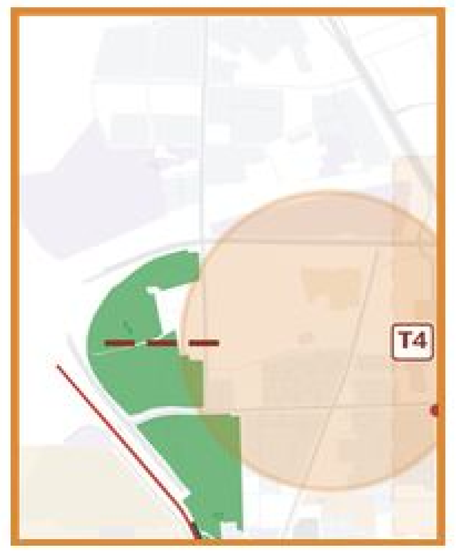


01 Near-campus innovation translation network
02 Station exchange node
03 Campus boundary gradient
04 Everyday living base for talent
05 Low-disturbance incremental renewal

STRATEGIC FRAMEWORK + CONCEPTUAL SPATIAL CONTROL FOR THE T2/T3 RESPONSE WINDOW

C | Zhongzhiyuan AI Autonomous Innovation Acceleration Area

About 192.1 ha | full-stack autonomous AI innovation system RESPONSE WINDOW T4



01 Full-stack AI industrial base
02 Public innovation interface
03 Integration of buildings, green space and water
04 Industry display and public exchange network
05 External transport and green interfaces

STRATEGIC FRAMEWORK + CONCEPTUAL SPATIAL CONTROL FOR THE T4 RESPONSE WINDOW

Why the three areas differ in design depth

Dazhongsi is the only place along the corridor where several problem types overlap in the same segment and where the factual record is most complete; it is therefore taken to object level, with a partial plan and a key section. The other two areas are answered with a strategic framework plus conceptual spatial control at the response window: no precise parcel placement, no building scale, no retain/renovate/demolish judgement and no statutory land-use change.

AI Scenarios, Delivery and Operation

Scenario — space — operation: where, by which mode, in what sequence

Delivery is not an investment schedule, but a question of which mode applies to which segment.

Twelve scenario cards (three of them industry test-and-verification scenarios), three landmarks upgraded from existing urban relation nodes, four delivery modes and a three-phase sequence, all on the same spine and the same set of positions.

Corridor-wide scenario — space — operation network

S09 / S10 / S11 are the three industry test-and-verification scenarios (red). All are CONCEPTUAL PROPOSAL / CONDITIONAL; none has obtained data permission; test areas would be publicly notified and manual takeover is always possible.

Four delivery modes | which mode applies where

	T5 Beijing North	T1 Dazhongsi	T2 Wudaokou	T3 Tsinghua East–Xueyuan Road	T4 Zhongzhiyuan
DIRECT DESIGN	Gateway arrival organisation at the station front; east–west link organised through the station area	Extend the entrance-to-spine arrival sequence; organise publicness in the North Third Ring bridge zone	Fill the western access gap (small scale, operation first)	Strengthen publicness at existing crossings	
PLANNING CONTROL	Reserve the Line 13A interface; conditional conversion of the shed land east of the yard	Controlled reconstruction of the renewal parcel and the Lanjing interface	Write interface conditions into the Wudaokou station rebuild	Conditional controlled exchange openings (agreed with the universities)	Conditional green-network interface, northern end–Qinghe
MANAGEMENT / OPERATION		Arrival hours and programme organisation in the 1733 forecourt	Activation of the start-up area · time-shared use		A continuous public interface between industry, green space and everyday space
KEEP / NO DESIGN				KEEP EXISTING CONDITIONS by default: no fence removal, no forced opening	If the crossing condition does not hold → keep existing conditions

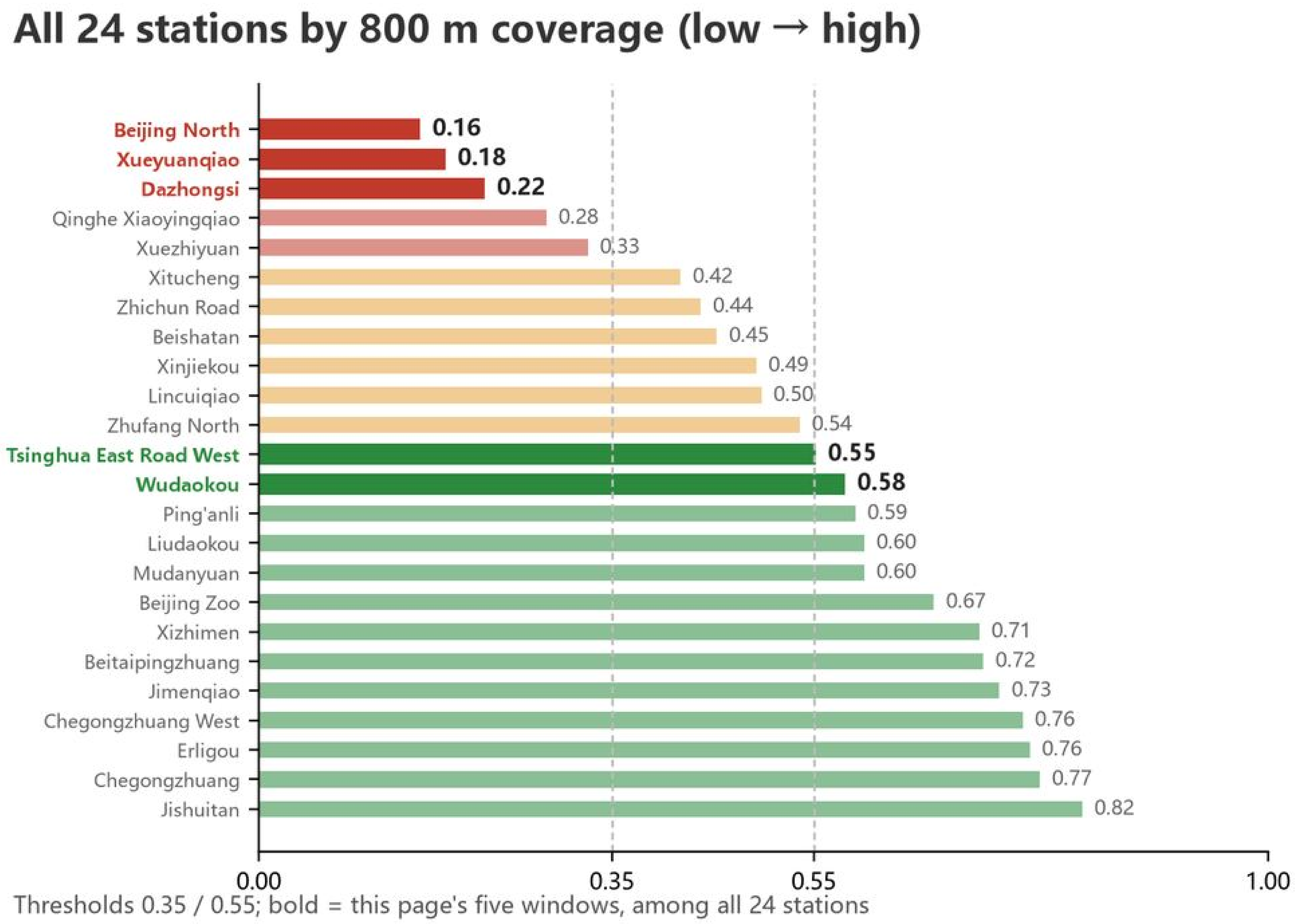
Three-phase sequence

The phasing is a conceptual sequence ordered by the priority of relation actions. It is not a development programme, an investment schedule or an approval timetable. No cost, staffing, investment figure, quantity of works or responsible body is proposed; all operating bodies are suggested role categories.

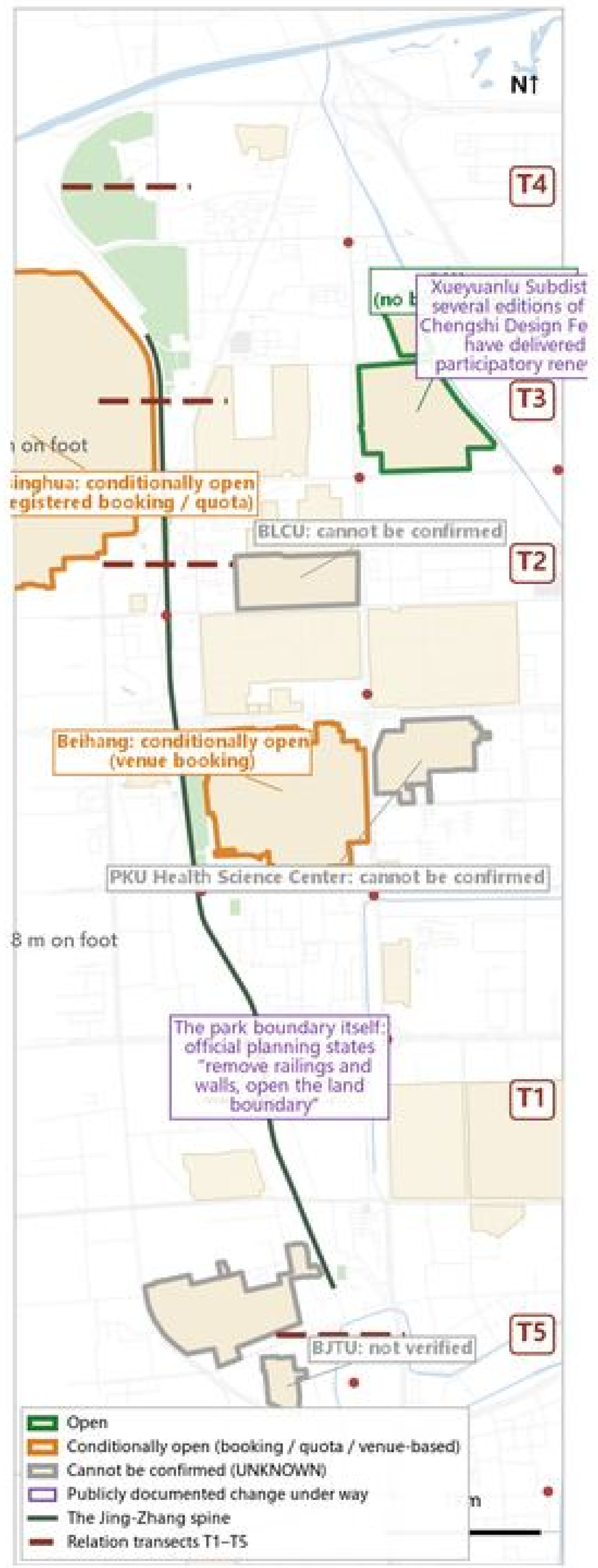
Metrics and compliance begin by stating what evidence each number rests on.

This board is not an audit back end. It gives the verifiable evidence first, then three verifiable diagnoses, and finally states what is verified, what is design-defined and what remains to be verified.

Strongest verifiable evidence



All 24 stations ranked by 800 m coverage | low values concentrate in the southern section and at bridge-zone stations.



Campus openness | Open / Conditionally open / Cannot be confirmed.

Residential node → nearest registered park entrance | green ≤400 m / amber ≤1,200 m / red >1,200 m.

Three Verifiable Diagnoses

01 | Real access to stations

800 m real coverage across 24 stations: 0.16–0.82

Beijing North 0.16 · Xueyuanqiao 0.18 · Dazhongsi 0.22

A Station ≠ Real Access

02 | Proximity-to-access gap

About 97 m to the park edge / about 700 m to the nearest entrance

Walked verification on the Zhichun Road segment (same stretch, same network)

Spatial Proximity ≠ Easy Access — a gap of about 7.2×

03 | Ecological continuity to public use

Five structural breaks, four of them linked to grey infrastructure

Residential node to park entrance: walked / straight-line ratio about 1.4–3.7× (sample)

Ecological Continuity ≠ Continuous Public Use

Plot ratio, building height control and the number of retain/renovate/demolish parcels remain undetermined: the official regulatory plan and redline are not public, and no design-model value is substituted for a statutory indicator.

Status boundaries

VERIFIED

800 m network coverage per station (own indicator) · classification of the seven existing crossings · park entrances and share of negative interface · plan-level factual relations at T1 (1733 / Entrances A-B / station structure / Line 13 / Jing-Zhang / North Third Ring)

DESIGN-DEFINED

Forecourt arrival · public hinge within the station · upper-level public transfer · heritage park interface · opening locations of the conditional controlled exchange interface

TO VERIFY

Exact levels · vertical circulation within the station structure · engineering interfaces · landing conditions of N2 · headroom and structure under the bridge · opening hours and rules of each university

NOT PROPOSED AT THIS STAGE

Any unverified new rail crossing · statutory land-use change · retain/renovate/demolish judgement · plot ratio and building height control

Compliance coverage

- ✓ **Three-scale scope** A0-01
- ✓ **Three key areas** A0-05
- ✓ **Mobility and blue–green** A0-03 / A0-04
- ✓ **AI scenarios** A0-06
- ✓ **Delivery and operation** A0-06
- ✓ **Metrics recomputation** A0-07

Evidence for the three diagnoses is on the left of this board and on A0-03 / A0-04. Once the official redline and the formal boundaries of the three key areas are released, the extents and statistics must be recomputed in full.

Provisional submission extent: about 11.41 km². This is the extent recorded in the submission package and is not a design-performance indicator.